

Northwind Platform — Architecture Overview

Services

The Northwind platform is composed of five core services. The Orders Service accepts customer orders and is the system of record for order state. It calls the Inventory Service to reserve stock and the Payments Service to charge the customer. The Payments Service integrates with Stripe as the payment processor. The Notifications Service sends transactional email through SendGrid and SMS through Twilio.

Data stores

The Orders Service stores order data in a PostgreSQL database named `orders_db`. The Inventory Service uses a separate PostgreSQL database named `inventory_db`. Both services publish domain events to an Apache Kafka cluster on the topic `'northwind.events'`. The Notifications Service consumes those events to decide when to message customers.

Authentication

All services authenticate requests using JWT access tokens issued by the Auth Service. Access tokens expire after 15 minutes; refresh tokens expire after 30 days. The Auth Service is backed by Redis for token revocation lists.

Deployment

Northwind runs on Kubernetes in the AWS us-east-1 region. Each service is deployed as a separate Deployment with a minimum of 3 replicas. Container images are built by GitHub Actions and pushed to Amazon ECR. Production deploys require approval from a member of the Platform team.